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Optimization of a *Chlorella vulgaris*-Based Carbon Sequestration Technique Using an Alkaline Medium of Wood Biomass Ash Extract

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Abstract: Algae communities as primary producers are essential elements of aquatic ecosystems and contribute significantly to oxygen production, carbon dioxide fixation, and nutrient transport processes in water bodies. The use of algae-based carbon capture and storage technologies does not produce harmful by-products that require disposal, and the resulting algal biomass can be valuable across various industrial sectors. In this study, model experiments were conducted to develop sequential absorption–microalgae hybrid CO₂-capture methods. To facilitate CO₂ capture from flue gases, wood biomass ash (WBA), an agricultural by-product, was utilized for its alkaline properties, while the flue gas scrubbing medium was regenerated by algae that restored alkalinity during their growth. In our experiments, one of our goals was to determine the optimal conditions for achieving maximum algal biomass growth in the shortest possible time. The suitability of WBA for flue gas cleaning was tested via simulation of CO₂ introduction. Moreover, a method was developed to determine the dissolved inorganic carbon content with the use of an OxiTop device monitoring the changes in pressure. The applied device was a closed, static, and pressure-based respirometer originally designed to determine the biological activity of microorganisms in both solid and liquid samples. In addition, the effects of CO₂-enriched WBA extract on algae cultivation were also analyzed, confirming that it imposed no growth inhibition and identifying the concentration (10% WBA) that optimally promoted algal growth. The optimal initial algal concentration and nutrient conditions for maximum growth were also determined.

Keywords: carbon capture; carbon utilization; algal biofuel; by-product utilization; dissolved inorganic carbon; wood biomass ash; microalgae; alkalinity; algae biomass



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1. Introduction

Global carbon dioxide (CO₂) emissions have been a critical environmental issue for decades, driven largely by human activities such as the burning of fossil fuels (e.g., coal, oil, and natural gas) used for energy production, transportation, and industrial processes. Additional sources include agriculture, waste disposal, garbage incineration, residential sector, and mining [1]. The mean global concentration of CO₂ in the atmosphere had increased from 280 ppm in the 1700s to 380 ppm in 2005 [2] and to 427 ppm by the year 2023 [3]. CO₂ has a long atmospheric lifetime of 50 to 200 years [4]. However, only about 45% of emitted CO₂ remains in the atmosphere [5], while the rest is absorbed by the ocean (25%) [6] and by terrestrial ecosystems such as forests (30%) [7]. This balance can fluctuate annually due to natural climate cycles and other influencing factors [8]. Deforestation also

contributes to increased CO₂ levels in the atmosphere, since forests are significant factors in CO₂ absorption [9,10]. The emission of primary greenhouse gases (CO₂, CH₄, and N₂O), is a key driver of climate change, since greenhouse gases can trap and retain thermal radiation in the Earth's atmosphere, leading to global warming [1].

The wide scale of emission sources highlights the importance of adopting strategies to reduce greenhouse gas emissions. These strategies are essential for addressing climate change and advancing sustainable development. International efforts to reduce emissions, especially the Kyoto Protocol in 1997 and the Paris Agreement in 2015, play a decisive role in forming a collective front against climate change [11]. Meanwhile, the European Union has adopted a net emissions reduction target of 55% by 2030. This objective sets the foundation for the EU to attain climate neutrality by 2050 [12]. While some countries and sectors have made efforts to reduce their carbon footprint through renewable energy (including wind, solar, and hydropower), energy efficiency, carbon capture technologies, and reforestation, global CO₂ emissions continue to grow, particularly in developing economies, posing a major challenge for climate action. Total emissions amounted to 38.52 gigatons of CO₂ equivalents (GtCO₂eq) in 2022, highlighting the urgent need for collective efforts to mitigate the adverse effects of climate change [1]. Global CO₂ emissions from industry and fossil fuels were 37.15 billion metric tons (GtCO₂) in 2022. It is expected that emissions will increase by 1.1% in 2023 to a record high of 37.55 GtCO₂ [13]. According to the Emissions Database for Global Atmospheric Research, the power industry is the highest contributor, with 38.1% of overall emissions, followed by transport (20.7%) and industrial combustion (17%) [1,14]. From the early 1970s to 2022, there was a consistent trend of increasing emissions of greenhouse gases (CO₂ represents 73.5% of the total emissions). However, a temporary reduction in CO₂ emissions was observed due to the COVID-19 pandemic in 2020, although a new increase was observed from 2021 [15–17]. The ongoing increase in atmospheric CO₂ levels presents serious risks including biodiversity loss, rising sea levels, and more extreme weather events. The observed continuously increasing tendency indicates the urgent need for global collaboration and effective measures to decrease emissions of CO₂ and other greenhouse gases, to mitigate the adverse effects of anthropogenic activities on the global climate and environment [1].

Based on the previous 'European Green Deal' and the 'European Climate Law', the production and utilization of renewable energy sources have also been emphasized [12,18–20]. Notably, bioenergy remains the predominant source of renewable energy within the EU, despite the significant growth of wind and solar energy in the past decade. Biomass for energy purposes accounts for nearly 60% of renewable energy sources in the EU. In 2017, forest biomass accounted for 69% of the total biomass used for energy production [21,22]. According to global estimates, 900–1000 million tons of energy waste are generated annually, with 476 million tons of ash from biomass combustion. In the EU, this amount reaches 100 million tons per year. Consequently, it seems reasonable to reuse the ash from biomass combustion for various applications, including agriculture, remediation of degraded land, environmental protection, zeolite synthesis, recovery of rare earth metals, and plastic production [23]. The utilization of cement kiln dust (CKD) in carbon capture technologies has long been a focus of research [24–26], due to the well-documented and significant contribution of the cement industry to global CO₂ emissions. However, the quality of CKD can vary significantly based on the combusted materials. Therefore, the application of wood biomass ash (WBA), which can be more precisely qualitatively characterized, has emerged as a favorable alternative [27]. This by-product serves as a valuable fertilizer component, as WBA is rich in essential nutrients (e.g., calcium (Ca), potassium (K), and microelements). However, it may also contain environmentally harmful substances, including heavy metals and combustion-derived substances, such as polycyclic aromatic hydrocarbons (PAHs) and volatile organic compounds (VOCs) [23,28].

Our approach promotes the use of WBA as agricultural waste to facilitate CO₂ capture through its alkaline properties, while regenerating the flue gas scrubber medium via algae, which re-alkalizes the medium during its growth. The increase in pH is attributed to

the external conversion of HCO_3^- into CO_2 and OH^- during photosynthesis, facilitated by algal cells with a high affinity for CO_2 uptake [29]. Microalgae are highly efficient at CO_2 fixation, absorbing considerably more CO_2 per area than terrestrial plants, which are limited by soil, water, and climate requirements. Controlled microalgae cultures can minimize moisture loss, leading to higher photosynthetic efficiency and enhanced CO_2 absorption [30]. Carbon rarely limits algal growth under natural conditions [31], but the low atmospheric CO_2 concentration ($\sim 0.042\%$) is insufficient for intensive algal cultivation. To overcome this limitation, artificial carbon sources such as NaHCO_3 or compressed CO_2 are usually added, although these can be costly [32]. Therefore, the current research focuses on cost-effective, sustainable sources of CO_2 , such as power plant flue gas, which contains up to 15% CO_2 [33–35], and biogas from anaerobic digestion [36–38].

Algal communities in aquatic ecosystems serve as primary producers, constitute the majority of biomass, and play an important role in the carbon cycle and oxygen production. Moreover, they are essential elements of aquatic food webs and nutrient transport processes [30,31]. Due to their fast growth and sensitivity to changes in water quality, algae are often used as indicator organisms during ecotoxicological testing to evaluate the possible environmental risks arising from water contaminants. In addition to the crucial importance of algae in surface waters, algal communities play a multifaceted role in the sustainable management of water resources. Algae contribute significantly to the maintenance and improvement of water quality. They are natural biofilters capable of absorbing, removing, and accumulating nutrients and certain water pollutants (e.g., phenols, pesticide active ingredients, and hydrocarbons). Therefore, algal species offer a sustainable approach to water remediation. However, the massive proliferation of certain species of algae can lead to eutrophication of surface waters, resulting in the deterioration of water quality. The potential of algae in the water–energy nexus is increasingly being recognized, as they offer a renewable resource for biofuel production. Algae play a crucial role in carbon sequestration by absorbing carbon dioxide from the atmosphere and incorporating it into their biomass [39]. Thus, the application of microalgae-based carbon capture and storage (CCS) technology can contribute to mitigating climate change.

The effects of elevated CO_2 concentrations on the growth and viability of several microalgae strains, including *Arthrospira platensis*, *Chlorella ellipsoidea*, *C. vulgaris*, *Gloeotila pulchra*, and *Elliptochloris subsphaerica*, were examined. Cultures were grown at constant CO_2 levels (0.04%, 3%, 6%, and 9%), and the highest biomass growth rate ($0.37 \text{ g L}^{-1} \text{ day}^{-1}$) was observed for *Chlorella vulgaris* at 3% CO_2 [39]. The impact of elevated CO_2 levels on algal growth varied. A 95% increase in dry weight was observed for *C. vulgaris* with 1% CO_2 -enriched air compared with the control, supporting the application of high CO_2 levels for maximum productivity [40]. However, under extreme CO_2 levels (40%), extracellular carbonic anhydrase (CA) activity decreased significantly, with almost no intracellular activity [41]. CO_2 fixation rates of 0.93 and $0.69 \text{ g L}^{-1} \text{ day}^{-1}$ were achieved with 2 and $1 \text{ g L}^{-1} \text{ NaHCO}_3$ in 100 mL flasks, respectively [42]. CO_2 bubbling resulted in significantly higher growth rates, while aerated cultures reached a value of $0.045\text{--}0.075 \text{ g L}^{-1}$. The application of model flue gas with 15% CO_2 resulted in a growth rate that was nearly ten times higher ($0.624 \text{ g L}^{-1} \text{ day}^{-1}$) [43]. Nevertheless, lipid content and biomass productivity decreased as CO_2 concentration increased (2–15% CO_2) [44]. Conversely, *C. vulgaris* adapted rapidly to 5–12% CO_2 levels with no effects on photosynthesis [45], while growth of *C. vulgaris* was indicated at 18% CO_2 , suggesting its potential for CO_2 mitigation in flue gas emissions [46]. Additionally, CO_2 supplementation can prevent pH inhibition in carbon-limited cultures [31,47]. According to another study, a membrane photobioreactor fixed CO_2 0.95–5.40 times more efficiently than conventional bioreactors [48]. The proposed system used a sequential method, where the CO_2 scrubbing and solution regeneration through the algal carbon fixation phase were separated.

Microalgae-based CCS technology presents a promising solution, as microalgae are estimated to produce about half of atmospheric oxygen while simultaneously using CO_2 for photoautotrophic growth [49]. Microalgae-based CCS strategies offer several advantages.

Microalgae grow faster and fix CO₂ more effectively than conventional plants, and completely recycle CO₂ by converting it into chemical energy via photosynthesis [50,51]. Energy production from microalgal biomass through biochemical, thermochemical, chemical, and direct combustion processes yields a variety of end products, such as bioethanol from fermentation, biomethane from anaerobic digestion, syngas from gasification, biodiesel, bio-oil, and charcoal from pyrolysis [43,52]. Numerous studies, especially on biodiesel production, have focused on developing cost-effective, large-scale production methods capable of competing with petroleum-based fuels. Although third-generation fuels derived from algae offer significant advantages over traditional crop-based fuels, the release of CO₂ back into the atmosphere during combustion means that biomass energy cannot be considered a form of carbon storage technology [53]. Microalgae provide a sustainable feed alternative for livestock. In addition, microalgae help to balance the needs of food, feed, and biofuel while lowering environmental impact [54]. Biotechnological advancements have optimized microalgal strains to efficiently produce nutrients and sequester carbon, enhancing their roles as “cell factories” [55]. However, further research on risk assessment is essential for the safe utilization of microalgal biomass derived from industrial flue gases [56]. Although the CO₂ captured by microalgae is eventually released back into the atmosphere when used in fuel or feed, microalgae-based chemical production can contribute to negative emissions. Moreover, land freed from conventional agriculture could be used for afforestation, further reducing CO₂ emissions [57].

Summarizing our objectives, our approach advocates the use of WBA as agricultural waste to facilitate CO₂ capture through its alkaline properties while regenerating the flue gas scrubber medium via algae, which re-alkalizes the medium during its growth. Our primary objective was to contribute to the implementation of this industrial technology through laboratory-scale model experiments. In these experiments, we aimed to identify the most optimal settings for maximizing algal biomass growth in the shortest possible time. Firstly, the suitability of WBA for flue gas cleaning was tested with a simulation of CO₂ introduction with a gas cylinder dispersing fine gas bubbles into the WBA extract via a diffuser. Then, the total dissolved inorganic carbon content of the extract was determined using OxiTop respirometer devices (see Section 2.3) to monitor the change in pressure. Furthermore, the potential of CO₂-enriched WBA extract for algal cultivation was also analyzed. The possible inhibitory effects of the washing solution on the algae, as well as the WBA concentration, initial algal concentration, and nutrient conditions that most optimally supported growth, were identified.

2. Materials and Methods

2.1. Characterization of the Applied Strain of Algae and the Culturing Conditions

During our modeling experiments, a stock culture of *C. vulgaris* was used as a test organism. The applied unicellular green algal strain (*Chlorella vulgaris* Beijerinck—CCAP 211/11b) was sourced from the Culture Collection of Algae at Charles University in Prague (Prague, Czech Republic) and cultured in Scharlau Algae Broth, a nutrient salt medium (Scharlab group, Barcelona, Spain) [58]. The application of *C. vulgaris* is advantageous due to its resilience to environmental fluctuations, tolerance to high CO₂ levels during cultivation, and ease of handling. Additionally, *C. vulgaris* was selected because of its extensive scientific documentation and high market demand [59]. Following the manufacturer’s instructions (Scharlab group, Barcelona, Spain), 1.87 g of the nutrient mixture was added to 1 L of the algal suspension. *C. vulgaris* was cultured in Erlenmeyer flasks at a controlled temperature (25 ± 1 °C) and under continuous illumination provided by four Osram Biolux full-spectrum fluorescent tubes (T8 L30W/965 Daylight 6500K, 26 × 895 mm) emitting 2665 lux ($59.22 \mu\text{mol}/\text{m}^2/\text{s}$) at the surface of the glass shelf [58,60].

2.2. Extraction of Wood Biomass Ash

The WBA sample used in the experiment was collected from fly ash derived from a biomass power plant located in Szakoly, Hungary. The power plant uses locally sourced

wood-based biomass that is stored and chopped on site for combustion. The fuel enters the 50 m³ bunker in front of the boiler through a screen and an iron magnet. The capacity of the unit is 19.8 MW, its efficiency is 30.7%, and the annual operating time is 7900 h. The composition of the biomass changes seasonally, which affects the properties of the ash depending on the raw material. The properties of the ash are also influenced by the duration of combustion, the temperature, the cooling time, and the nature of the initial material [61,62]. In the laboratory, a 1:10 distilled water extract was prepared from the WBA sample in three replicates (Table 1).

Table 1. Composition of the 1:10 distilled water extract of wood biomass ash (WBA).

Element	Concentration (mg L ⁻¹)
K	2233.23 ± 67.18
Ca	546.57 ± 34.53
Al	4.30 ± 1.05
Fe	1.15 ± 0.28
Zn	0.10 ± 0.04
Cu	0.06 ± 0.01

The selected elements in the extracts were determined with an atomic emission spectrometer (MP AES, Agilent 4210, Santa Clara, CA, USA). The high contents of potassium (K) and calcium (Ca) measured in the extract were consistent with previously reported high concentration values [23,28]. The elemental analysis was performed in triplicate.

After 24 h of mixing, the solution was filtered through filter paper (Whatman No. 42, pore size = 0.25 µm). The total alkalinity of the extract was determined using an automatic potentiometric titrator (Auto Titrator SCHOTT Instrument TitroLine easy Germany), in three replicates. The transition point was determined using Origin software (version number 6.0) and differential function fitting (Figure 1).

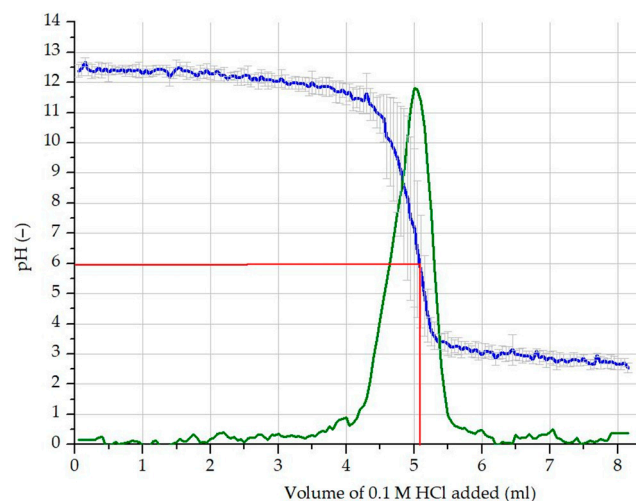


Figure 1. Determination of total alkalinity by automatic potentiometric titration: average measured pH (blue line); the first derivate curve of the measured pH (green line); volume and pH values corresponding to the inflection point of the measured pH and the maximum of the first derivative curve, pH = 6.01, added acid = 5.1 mL (red lines).

For the experiment, 5.1 mL of acid was added to 10 mL extract. The concentration of [OH⁻] in the WBA was therefore calculated to be 0.51 mol kg⁻¹, equivalent to 0.051 mol L⁻¹ in the 1:10 water extract, with a pH value of 12.71.

2.3. Model Experiment for the Optimization of Wood Biomass Ash Extract

After applying the extraction method described in the previous chapter (Section 2.2), the effects of different treatments with various rates of WBA (0%, 10%, 25%, and 50%) were assessed along with a blank unit. The exact compositions of the treatment and the blank are summarized in Table 2. To ensure that the growth results could be compared with the optimal growth, treatments and control experiments were conducted under the same laboratory conditions using conventional nutrient media (Scharlau Algae Broth, Scharlab group, Barcelona, Spain). The bubbled CO₂ gas from a cylinder passed through the samples with the use of a diffuser (ISTA Ceramic UFO CO₂ Diffuser, 25 mm Suction Cup). After the CO₂ scrubbing process, the pH of all samples was adjusted from their initial pH to 6±0.05 (presented in Table 2). The samples were placed in 500 mL Erlenmeyer flasks on the illuminated glass shelf, where the pH was monitored every 12 h during the cultivation using a digital pH/mV meter (OP-211/2, Radelkis Kft., Budapest, Hungary) [63]. The effects of the treatments as well as the control conditions were investigated in three repetitions. Algal biomass was assessed by measuring the optical density at 440 nm using a spectrophotometer (Jenway 6405 UV/VIS, Cole-Parmer Instrument Co., Neots, UK). Then, the concentration of biomass was calculated according to the following formula (1) [58]:

$$\text{Algal biomass in mg (dry matter)} = \ln(1 - \text{OD}(440 \text{ nm})/3.29)/(-0.0022) \quad (1)$$

Table 2. Experimental design and composition of the treatments and blank units during the model experiment for wood biomass ash extract optimization.

Treatment	WBA Extract ¹ (mL L ⁻¹)	Algae Broth ² (g L ⁻¹)	BDW ³ (mL)
blank	-	1.87	500
0%	-	1.87	500
10%	100	1.87	450
25%	250	1.87	375
50%	500	1.87	250

Note(s): ¹ wood biomass ash extract, ² nutrient salt media for algae, ³ boiled distilled water.

Total dissolved inorganic carbon (DIC) is usually measured by acidifying a seawater sample to convert HCO₃⁻ and CO₃²⁻ into undissociated CO₂ for extraction and titration [64]. In our study, we utilized OxiTop devices to determine DIC based on pressure measurements, the developed method not having previously been documented in the literature. An OxiTop device (OxiTop-IDS B6M-2.5 WTW, Weilheim, Germany) is a closed, static, pressure-based respirometer originally designed to assess the biological activity of microorganisms in both solid and liquid samples [65]. During the measurement, a CO₂-enriched algal suspension was sealed in the OxiTop chamber with a beaker containing 50 mL of 1 M HCl. After measuring the base pressure for 10 min, the acid was mixed into the suspension by shaking; thus, CO₂ was released. The final pressure stabilized over the next 20 min, and the results were then obtained. Before the model experiment, the CO₂-diffuse solution containing algae and ash extract was divided: half was used for the OxiTop measurement, and the other half was used for the algal cultivation model. After the experiment was completed, the DIC values of the solutions were also measured in triplicate, providing both initial and final DIC values for each treatment. Based on the measured pressure, the molar quantity of CO₂ was calculated via the ideal gas law to determine DIC values according to the mass of carbon in a molar quantity of CO₂ (12 g mol⁻¹) and the volume of the suspension (200 mL).

Our method was adapted from the Scheibler calcimeter according to the Hungarian Standard MSZ-08 0206-2:1978 [66], which measures soil carbonate content by generating CO₂ gas through the addition of acid, resulting in displacement in a measuring cylinder [67]. In contrast to this volume-based approach [68], our OxiTop device detected

pressure changes in a thermostatic setup maintained at a constant temperature of 20 °C. If a standardized acid mixing method is developed, this device could be also used to investigate the kinetics of CO₂ release.

2.4. Experiment for the Optimization of the Initial Algal Biomass and Nutrients

During the optimization of initial algal biomass and nutrient content, the experimental settings were similar to those for the WBA extract optimization (Section 2.3). The exact compositions of the applied treatments and the blank are summarized in Table 3. The effects of the treatments and also the control conditions were investigated in three repetitions. At the beginning and end of the test, the solutions were centrifuged at 5000 rpm for 15 min (Hermle Z 206 A centrifuge, Hermle Labortechnik GmbH, Wehingen, Germany), followed by filtration through 0.45 µm filter paper to measure nutrient content. The N content was measured by Parnas–Wagner steam distillation, determining NH₄⁺-N and NO₃[−]-N values [69]. The P content was extracted using an ammonium lactate solution and determined with a spectrophotometer (Photometer Spekol 221, Iskra Elektronik, Stuttgart, Germany) [70]. The K content was determined with the use of a flame photometer (Flamom-B, Medicor Zrt., Budapest, Hungary) [71].

Table 3. Experimental design and composition of the treatments and blank units during the model experiment for initial algal biomass and nutrient optimization.

Treatment	WBA Extract ¹ (mL L ^{−1})	Algae (mL L ^{−1})	Algae (v/v %)	Algae Broth ² (g L ^{−1})	Algae Broth % ³
blank	0	20	2	0	0
A2N0	100	20	2	0	0
A2N100	100	20	2	1.87	100
A5N50	100	50	5	0.94	50
A8N0	100	80	8	0	0
A8N100	100	80	8	1.87	100

Note(s): ¹ wood biomass ash extract, ² nutrient salt medium for algae, ³ 100% = quantity recommended by the manufacturer.

2.5. Statistical Analysis

During our model experiments, the effects of the treatments according to the different experimental compositions were investigated in three repetitions. In addition, the control conditions were tested also in triplicate. The statistical analyses were performed with the use of the R Statistical program 4.2.1. (R Development Core Team, Vienna, Austria). Before the statistical analysis, the normality of the data and the homogeneity of variance were checked by Shapiro–Wilk and Levene’s or Bartlett’s tests at a significance level of 0.050. The effects of the treatments were evaluated using one-way ANOVA during the model experiment to optimize the WBA extract. During the experiments for the optimization of the initial algal biomass and Algae Broth nutrient level, the effects of the two factors (initial level of algal concentration and Algae Broth nutrients) and their possible interaction were investigated with the use of two-way ANOVA analysis. Tukey’s honest significant difference (HSD) testing was used for post hoc analysis to assess the significant differences between groups. The data were evaluated using the Kruskal–Wallis test if the conditions for ANOVA were not met, with the use of Dunn’s test for the comparison of the different groups at a significance level of 0.050. Where the initial and final values were available, the statistical analysis of those parameters (algal concentration, DIC, N, P, and K concentrations) was also carried out with the use of the R Statistical program 4.2.1, according to the applied treatments. For each treatment, the differences between the initial and final values were evaluated by two-sample *t*-tests.

3. Results

3.1. Results of the Wood Biomass Ash Extract Optimization

During our experiment, the applied culturing media were originally basic due to the addition of WBA. With the addition of a higher rate of WBA, the media became more alkaline, and thus, more CO₂ was captured. The highest final pH value was observed for the 10% WBA treatment (Figure 2). A sigmoidal curve was fitted to the average values of measured biomass data across all treatments ($R^2 \geq 0.992$), as shown in Figure 3. The analysis of these functions enabled us to determine the theoretical maximum producible biomass under these conditions (A2) as well as the time required for that growth (dx) (Table 4). It was noted that the maximum biomass was higher in the 25% treatment (176 mg L⁻¹ dry mass), while the time required for growth was more favorable in the 10% treatment.

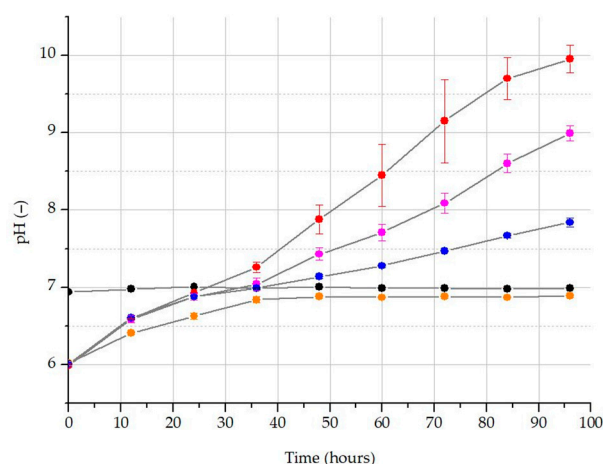


Figure 2. Changes in the pH value of differently carbonated treatments: blank treatment without CO₂ diffusion (black); 0% WBA addition with CO₂ scrubbing (orange); 10% WBA addition with CO₂ scrubbing (red); 25% WBA addition with CO₂ scrubbing (purple); 50% WBA addition with CO₂ scrubbing (blue).

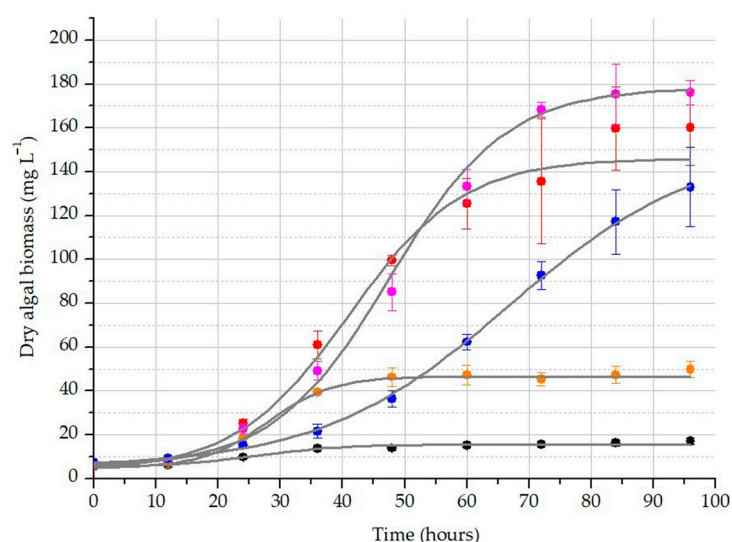


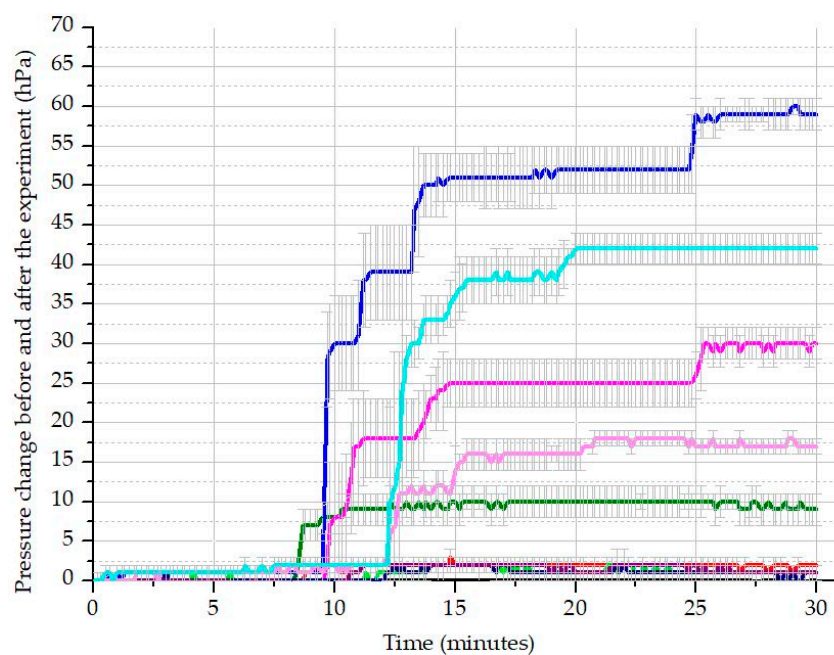
Figure 3. The growth of algal biomass in the differently carbonated treatments: blank treatment without CO₂ diffusion (black); 0% WBA addition with CO₂ scrubbing (orange); 10% WBA addition with CO₂ scrubbing (red); 25% WBA addition with CO₂ scrubbing (purple); 50% WBA addition with CO₂ scrubbing (blue).

Table 4. The determined parameters based on the fitted sigmoidal curves during the wood biomass ash extract optimization.

Treatment	Chi ² ¹	R ² ²	A1 ³	A2 ⁴	x0 ⁵	dx ⁶
blank	0.832	0.992	5.2	15.6	25.8	7.1
0%	0.359	1.000	4.9	46.6	27.9	5.2
10%	1.621	0.996	3.6	146.0	41.0	9.2
25%	1.778	0.998	5.0	178.3	47.6	9.4
50%	1.126	0.992	5.4	152.5	66.8	15.3

Note(s): ¹ difference between the observed and expected data; ² accuracy of curve fitting; ³ theoretical (asymptotic) minimum initial biomass weight; ⁴ theoretical (asymptotic) maximum final biomass weight; ⁵ the steepest point of the curve (where the growth rate was the fastest at this time); ⁶ the slope of the curve (the time required to reach maximum growth).

The average pressure results from the OxiTop system, along with standard deviations, are shown in Figure 4. To calculate the CO₂ released following each treatment, the initial basic values (after 5 min) were subtracted from the values after acidification (after 30 min), according to the known volume of air space in the bottles and the constant temperature of 20 °C provided by the thermostat. Our findings demonstrate that DIC can be measured using this device. Based on the calculated DIC results, expressed as mg of carbon per liter of solution (mg C L⁻¹) (Figure 5, Table 5), the 10% WBA treatment proved to be the most effective, with 93% of the dissolved carbon utilized by the algae. This indicates that, in this treatment, the algae nearly completely regenerated the CO₂-scrubbing medium.

**Figure 4.** The measured pressure values after acidification before and after the algae-cultivating model experiment: blank initial (black), blank final (dark grey); 0% WBA initial (red), 0% WBA final (purple); 10% WBA initial (olive), 10% WBA final (green); 25% WBA initial (magenta, 25% WBA final (light magenta); 50% WBA initial (blue), 50% WBA final (light blue).

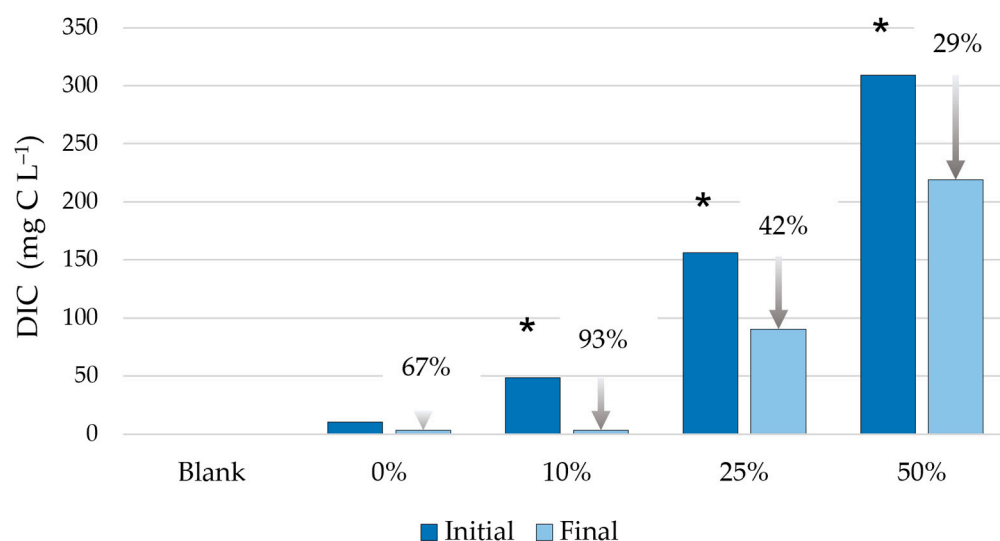


Figure 5. The measured total dissolved inorganic carbon (DIC) contents at the beginning and the end of the algae-cultivating model experiment (the percentages on the x-axis represent the concentration of the added WBA extract; * indicates significant difference between the initial and final DIC values at a significance level of 0.050).

Table 5. Summarized results of wood biomass ash treatments.

Parameter	Dimension	Blank	0% WBA ¹	10% WBA	25% WBA	50% WBA
initial algal conc. ²	mg L ⁻¹ (d.m.)	5.6 ± 0.5 a ³	5.5 ± 0.6 a	5.5 ± 0.8 a	6.5 ± 0.7 a	5.7 ± 0.8 a
final algal conc.	mg L ⁻¹ (d.m.)	* 17.1 ± 0.5 a	* 49.8 ± 3.6 b	* 160.2 ± 17.2 c	* 176.0 ± 5.6 cd	* 133.3 ± 18.0 d
change in biomass	mg L ⁻¹ (d.m.)	11.5 ± 1.6 a	44.3 ± 3.8 b	154.7 ± 17.1 c	169.5 ± 5.5 cd	125.8 ± 18.3 d
mass multiplication	-	3.10 ± 0.4 a	9.24 ± 1.3 a	29.32 ± 3.5 b	27.21 ± 1.8 b	17.83 ± 3.2 c
specific growth rate (μ) ⁴	day ⁻¹	0.28 ± 0.03 a	0.55 ± 0.04 b	0.84 ± 0.03 c	0.83 ± 0.02 c	0.72 ± 0.04 d
doubling time ⁵	days	1.97 ± 0.11 a	1.29 ± 0.07 b	0.86 ± 0.04 c	0.89 ± 0.02 c	1.02 ± 0.06 c
pH without CO ₂ addition	-	6.94 ± 0.01 a	6.79 ± 0.01 a	9.67 ± 0.20 b	11.18 ± 0.06 c	12.15 ± 0.04 d
change in pH	-	0.05 ± 0.01 a	0.87 ± 0.04 b	3.95 ± 0.17 c	2.98 ± 0.09 d	1.83 ± 0.06 e
initial DIC ⁶	mg C L ⁻¹	LOD ⁷	10.4 ± 0.00 a	48.6 ± 7.96 b	156.3 ± 9.03 c	309.2 ± 7.96 d
final DIC	mg C L ⁻¹	LOD ⁷	3.5 ± 3.01 a	* 3.5 ± 3.01 a	* 90.3 ± 3.01 b	* 218.9 ± 10.42 c
DIC removal	%	-	66.7 ± 28.9 ab	92.8 ± 6.5 a	42.2 ± 1.4 b	29.1 ± 5.15 b

Note(s): ¹ wood biomass ash extract; ² dry mass (d.m.); ³ different letters indicate statistically significant differences between the control and treated groups ($p \leq 0.05$); ⁴ specific growth rate (μ): $\mu = \ln(N_2/N_1)/(t_2 - t_1)$; ⁵ doubling time (t_d): $t_d = \ln(2/\mu)$ calculated using the specific growth rate formula [72,73]; ⁶ total dissolved inorganic carbon; ⁷ limit of detection; * indicates significant difference between the initial and final values ($p \leq 0.05$).

According to the statistical analysis of the results for the WBA treatments, higher final algal concentrations, biomass changes, and specific growth rates were detected for the WBA treatments compared with the control ($p \leq 0.038$). The highest final algal concentrations, biomass change, and specific growth rates were determined for 10% and 25% WBA treatments compared with the control ($p < 0.001$). Significant differences were demonstrated between the initial and final algal concentrations for each treatment ($p \leq 0.006$). Compared with the control, significantly higher pH values were measured for the 10%, 25%, and 50% WBA treatments ($p < 0.001$), while a significantly different pH change was demonstrated in each group ($p < 0.050$). The initial DIC values were significantly different in each treated group, while the highest final DIC values were measured for the treatment with 50% WBA ($p < 0.001$).

Significant differences were observed in the initial and final DIC values determined for treatments with 10%, 25%, and 50% WBA ($p \leq 0.003$). In contrast, there was no significant

difference between the initial and final DIC values measured in the treatment with 0% WBA ($p = 0.057$) (Figure 5). The highest DIC removal was observed for 10% WBA treatment ($p \leq 0.014$), although the difference was not significant compared with the 0% WBA treatment ($p = 0.223$). Based on the parameters relating to algal growth, the 10% and 25% WBA treatments were found to be the most appropriate, while the highest rate of DIC removal was detected for 0% and 10% WBA treatments (Table 5). Based on the results and tested parameters, the 10% WBA treatment can be considered the most effective.

3.2. Results for the Initial Algal Biomass and Nutrient Optimization

In the control sample, the increase in biomass led to a rapid rise in pH (Figure 6). Due to the lack of WBA extract, a significant amount of CO_2 was not captured in this sample, resulting in limited carbon available for algal growth. In the control, a very low buffering capacity was demonstrated. Consequently, the rapid formation of hydroxide ions associated with algal growth apparently caused a quick increase in pH in these samples. In contrast, WBA acted as a buffering system in the other treatments, rendering them significantly less responsive to algal activity. Regarding pH, three treatments (A8N100, A5N50, and A2N100) achieved similarly high levels.

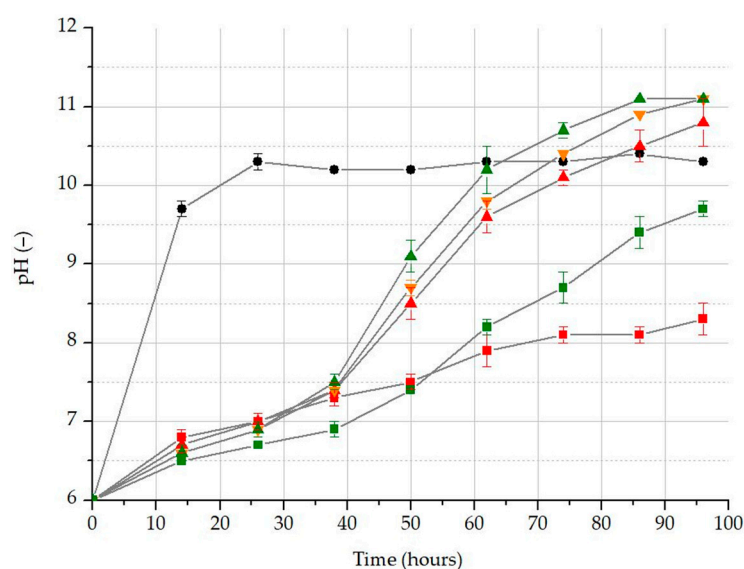


Figure 6. Changes in the pH values according to the initial algae biomass and nutrient levels: blank treatment (black dot); A2N0 (red square); A2N100 (red up triangle); A5N50 (orange down triangle); A8N0 (green square); A8N100 (green triangle).

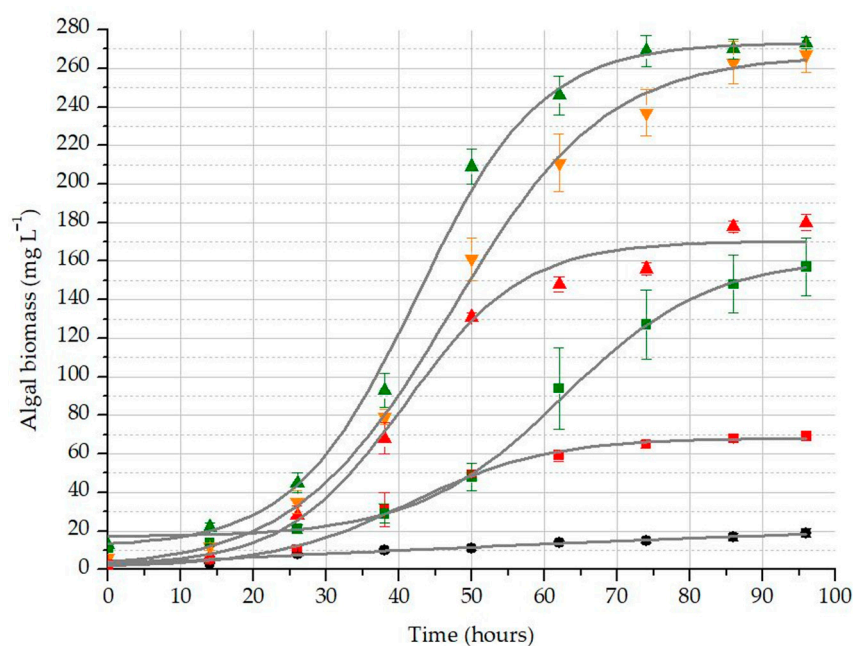
3.3. Results for Biomass Growth in the Differently Inoculated and Nutrient-Supplied Treatments

As shown in Table 6, A5N50 and A8N100 treatments were expected to reach their maximum around 270 mg L^{-1} (A2), while the lower dx value of the latter meant that it reached this maximum in a shorter time. The relatively high x_0 value for the A8N0 treatment suggests that it has greater reserves of available resources, as can also be deduced from the slightly flattened shape of its sigmoid curve (Figure 7). Interestingly, A8N100 treatment was associated with significantly faster growth than A2N100, suggesting that initial algal concentration may have had a stronger influence on biomass growth than nutrient supply.

Table 6. The determined parameters based on the fitted sigmoidal curves during the initial algal biomass and nutrient optimization.

Treatment	Chi ² ¹	R ² ²	A1 ³	A2 ⁴	x0 ⁵	dx ⁶
blank	0.515	0.971	n.a. ⁷	33.9	n.a. ⁷	97.2
A2N0	0.375	0.999	1.2	68.2	41.4	9.5
A2N100	7.767	0.996	2.0	170.3	40.8	8.1
A5N50	0.375	0.999	0.7	267.1	47.1	10.6
A8N0	0.082	0.997	17.0	161.6	62.4	10.0
A8N100	1.176	0.999	11.7	273.2	42.5	8.4

Note(s): ¹ difference between the observed and expected data; ² accuracy of curve fitting; ³ theoretical (asymptotic) minimum initial biomass weight; ⁴ theoretical (asymptotic) maximum final biomass weight; ⁵ the steepest point of the curve (where the growth rate was the fastest at this time); ⁶ the slope of the curve (the time required to reach maximum growth); ⁷ not applicable.

**Figure 7.** Growth of algal biomass in the differently inoculated and nutrient-supplied treatments: blank treatment (black dot); A2N0 (red square); A2N100 (red up triangle); A5N50 (orange down triangle); A8N0 (green square); A8N100 (green triangle).

3.4. Observed Changes in the Nutrient Content of the Media

The primary aim of analyzing each nutrient element was to evaluate the potential purification efficiency of algae when wastewater was used as a nutrient source. Interestingly, in the blank, A2N0, and A8N0 samples, the initial N content was almost minimal (Table 7), although biomass growth was generally observed (Figure 7, Table 8). However, the linear regression indicated that the correlation of the measured biomass increased with the consumption of N, P, and K nutrients. Relatively low correlation values were observed for P ($R^2 = 0.5569$), K ($R^2 = 0.5755$), and N ($R^2 = 0.7584$). The highest initial N level was detected for the A8N100 treatment ($p \leq 0.007$). Similarly, the lowest final N concentrations were also observed for the blank, A2N0, and A8N0 treatments. Additionally, the highest final N levels compared with the control were measured for A2N100 and A8N100 ($p < 0.001$). The statistical analysis suggested that the values of the initial and final N levels were significantly affected by the initial algal biomass and nutrient level ($p < 0.001$). In addition, the interaction between the two factors (initial algal biomass and nutrient level) was also significant with regard to the initial N values ($p = 0.003$). Compared with the initial N

concentrations, the level of N was significantly reduced in treatments where the initial level of Algae Broth nutrient was $\geq 50\%$ (A2N100, A5N50, and A8N100) ($p < 0.020$) (Table 7).

Table 7. The measured initial and final levels of nitrogen, phosphorus, and potassium in the media of the different treatments.

Treatment	N Concentration ¹ (mg L ⁻¹)		P Concentration ² (mg L ⁻¹)		K Concentration ³ (mg L ⁻¹)	
	Initial	Final	Initial	Final	Initial	Final
blank	3.7 ± 1.1 a ⁴	1.2 ± 0.2 a	34.7 ± 1.5 a	* 24.9 ± 1.7 ab	8.3 ± 2.2 a	* 0.0 ± 0.0 a
A2N0	3.2 ± 1.4 a	0.2 ± 0.1 a	30.6 ± 5.1 a	* 16.0 ± 5.6 b	242.3 ± 5.8 b	241.1 ± 8.6 b
A2N100	290.9 ± 4.0 b	* 256.9 ± 5.0 b	198.9 ± 10.1 b	* 36.4 ± 3.3 c	364.2 ± 6.8 c	* 342.2 ± 2.0 c
A5N50	168.8 ± 14.3 c	* 128.1 ± 11.5 c	124.8 ± 4.3 c	* 35.4 ± 1.0 c	327.9 ± 4.2 d	* 309.9 ± 6.7 d
A8N0	3.2 ± 0.0 a	1.0 ± 0.9 a	38.4 ± 1.1 a	32.2 ± 3.1 ac	257.3 ± 5.7 b	250.1 ± 4.6 b
A8N100	317.0 ± 8.9 d	* 254.6 ± 8.6 b	217.3 ± 40.7 b	* 23.2 ± 3.8 ab	404.1 ± 12.4 e	* 362.6 ± 10.9 e

Note(s): ¹ nitrogen concentration; ² phosphorus concentration; ³ potassium concentration; ⁴ different letters indicate statistically significant differences between the control and treated groups ($p \leq 0.05$); * indicates significant difference between the initial and final values ($p \leq 0.05$).

Table 8. Summarized results of initial algal inoculation and nutrient optimization in the various treatments.

Parameter	Dimension	Blank	A2N0	A2N100	A5N50	A8N0	A8N100
initial algal conc.	mg L ⁻¹ (d.m.) ¹	2.9 ± 0.49 a ²	2.4 ± 0.21 a	3.6 ± 1.18 a	6.0 ± 0.50 b	11.4 ± 0.88 c	13.2 ± 0.81 c
final algal conc.	mg L ⁻¹ (d.m.)	* 19 ± 1.6 a	* 69 ± 1.8 b	* 180 ± 3.8 c	* 267 ± 9.1 d	* 157 ± 15.0 e	* 273 ± 3.4 d
change in biomass	mg L ⁻¹ (d.m.)	16 ± 1.1 a	67 ± 1.7 b	177 ± 2.6 c	261 ± 8.9 d	146 ± 15.0 e	260 ± 4.0 d
mass multiplication	-	7 ± 0.6 a	29 ± 2.3 bc	54 ± 19.2 d	44 ± 3.1 bd	14 ± 1.3 ac	21 ± 1.4 ac
specific growth rate ³	day ⁻¹	0.47 ± 0.02 a	0.84 ± 0.02 b	0.98 ± 0.08 c	0.95 ± 0.02 c	0.66 ± 0.02 d	0.76 ± 0.02 bd
doubling time ⁴	days	1.46 ± 0.05 a	0.87 ± 0.02 bc	0.72 ± 0.08 d	0.75 ± 0.02 bd	1.11 ± 0.04 e	0.97 ± 0.02 c
change in pH	-	4.32 ± 0.02 a	2.27 ± 0.19 b	4.78 ± 0.27 c	5.11 ± 0.03 c	3.72 ± 0.12 d	5.06 ± 0.04 c

Note(s): ¹ dry mass (d.m.); ² different letters indicate statistically significant differences between the control and treated groups ($p \leq 0.05$); ³ specific growth rate (μ): $\mu = \ln(N_2/N_1)/(t_2 - t_1)$; ⁴ doubling time (t_d): $t_d = \ln(2/\mu)$; * indicates significant difference between the initial and final values ($p \leq 0.05$) [72,73].

Similar to the N levels, the initial P levels were also low in the blank, A2N0, and A8N0 treatments. Compared with the blank samples, the highest initial P concentrations were detected for A2N100 and A8N100 ($p < 0.001$). Compared with the control, the highest final P levels were measured for the A2N100 and A5N50 treatments ($p < 0.025$). In the other samples, significant differences compared with the blank were not detected in the final P levels ($p \geq 0.068$). The values of initial and final P levels were also significantly affected by the initial algal biomass and Algae Broth nutrient level ($p \leq 0.044$). Moreover, the interaction between the two factors also significantly affected the final P values ($p < 0.001$). A significant decrease between the initial and final measured P levels was observed for all treatments ($p \leq 0.029$), except the A8N0 treatment ($p = 0.061$) (Table 7).

Compared with the control, the highest initial and final K values were also detected for the A8N100 treatment ($p < 0.001$). Based on the results, the lowest initial and final K levels were determined in the blank samples ($p < 0.001$). In the samples of A2N0 and A8N0, there were no differences in the initial and final K concentrations ($p \geq 0.161$). The initial and final K levels were also significantly affected by the initial algal biomass and Algae Broth nutrient level ($p \leq 0.038$). However, the interaction between the two factors was not proved based on the statistical analysis of initial and final K levels ($p \geq 0.241$). Significant reductions in final K levels compared with initial K levels were not observed for the treatments where the initial level of Algae Broth nutrients was 0% (A2N0 and A8N0).

($p \geq 0.856$). In the other treatments, significantly decreased final K levels compared with the corresponding initial levels were detected ($p \leq 0.024$) (Table 7).

Compared with the initial algal concentrations, the final algal concentrations were significantly higher in each treatment ($p \leq 0.003$). Additionally, significantly higher final algal concentrations, biomass change, specific growth rates, and doubling times were detected for the different WBA treatments compared with the control samples ($p < 0.001$). The highest final algal concentrations and biomass change compared with the control were determined for the A5N50 and A8N100 treatments ($p < 0.001$). Additionally, the highest specific growth rates and the largest increase in biomass compared with the control were observed for the A2N100 and A5N50 treatments ($p < 0.001$) (Table 8), although the specific growth rates in these treatments ranked only within the top three for net biomass increase. The observed pH changes were not significantly different in those treatments where the initial level of Algae Broth nutrient was $\geq 50\%$ (A2N100, A5N50, and A8N100) ($p \geq 0.124$). In the other treatments, the pH change compared with the control was different in each group ($p \leq 0.020$). According to the statistical analysis, the measured and calculated parameters were significantly affected by the initial level of the Algae Broth nutrient ($p \leq 0.011$). The significant effect of initial algal concentration on the tested parameters was proven, except for the doubling time ($p = 0.134$). Moreover, the interaction between initial algal concentration and Algae Broth nutrient level was proven to be significant in the case of biomass multiplication ($p = 0.026$).

Comparing of the algal concentration and variable nutrient supplies with the resulting biomass increase (Figure 8), none of the parameters caused a significant growth above the average value. This phenomenon might be attributed to the depletion of another environmental resource, such as dissolved carbon, which could have acted as a limiting factor for growth. If this were not the case, the growth under the average parameter setting would ideally fall midway between the biomass levels observed under the two extreme parameter settings. If these two parameters were acting in opposition, nutrient availability appeared to exert a stronger influence on biomass. However, it is important to note that although both parameters can be scaled within the same dimension, comparability issues arise due to differences in scaling dimensions.

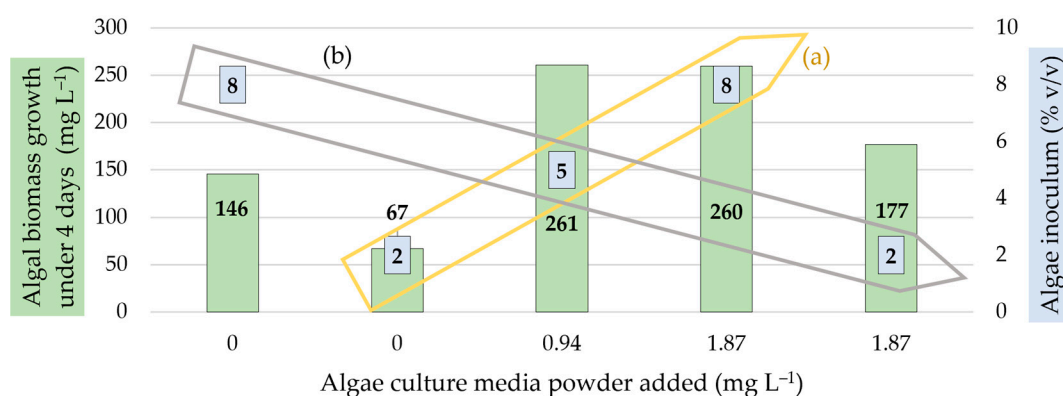


Figure 8. Trends indicated between the algal biomass growth obtained with increasing levels of nutrient media and the (a) directly and (b) inversely proportional initial algal inoculation treatments.

4. Discussion

Alkalization of the medium during algal biomass growth has been documented in various studies [29,74–76]. In our experiment, the culturing medium was initially basic due to the addition of WBA. As the WBA rate increased, the medium became more alkaline, leading to greater capture of CO₂. The highest final pH value was detected in the 10% WBA treatment (Figure 2), suggesting that algal growth in these groups was capable of fully regenerating the pH. One of the major findings of our research is the improvement of the low inhibitory effect of pH on the growth of *Chlorella vulgaris* compared with the effect of carbon limitation within the medium. According to Ihnken et al., *Chlorella* spp.

typically thrive at a pH range of 6 to 6.5 [77], while *C. vulgaris* can also grow effectively at an alkaline pH of 10.5. Additionally, the ability of these algae to tolerate extremely low pH is well documented, as their resilience underlies small-scale zooplankton control methods [32,78,79].

The applied OxiTop system is suitable for the measurement of DIC. During the WBA extract optimization, the 10% and 25% WBA treatments were identified as the most suitable according to the parameters for algal growth, whereas the highest DIC removal was observed in the 0% and 10% WBA treatments (Table 5). Overall, based on the results and evaluated parameters, the 10% WBA treatment emerges as the most effective option. According to the assessment of the initial algal biomass and nutrient optimization, the increase in biomass caused a rapid rise in pH in the control group (Figure 8). Due to the absence of significant CO₂ sequestration, limited carbon sources were available for algal growth. As a result, the control demonstrated very low buffering capacity. The rapid formation of hydroxide ions associated with algal growth is likely to have contributed to the quick increase in pH in these samples. In contrast, the presence of WBA in the other treatments acted as a buffering system, making them significantly less responsive to algal activity. The results of biomass growth in the differently inoculated and nutrient-supplied treatments suggest that initial algal concentration may have a greater impact on biomass growth than the availability of nutrients. According to the results, the values of biomass change, mass multiplication, specific growth rate, and doubling time measured for 10% WBA and A2N100 were consistent.

The main purpose of analyzing each nutrient element was to assess the potential purification efficiency of algae when utilizing wastewater as a nutrient source. Interestingly, in the blank, A2N0, and A8N0 samples, the initial N content was nearly negligible (Table 7), yet, biomass growth was generally observed. This phenomenon can possibly be attributed to the ability of certain algal species, such as *Chlorella*, to adjust their biochemical composition in N-deficient environments. Since N is primarily necessary for the construction of amino acids that form proteins, it is probable that the algae increased their lipid content at the expense of protein content [80,81]. This also explains why N deficiency did not inhibit overall mass growth. Notably, P is often considered as a limiting factor for the onset of eutrophication [82–84], but it was not depleted in any of the treatments, indicating that it was not a limiting factor in this experiment. It is important to note that this method combined with ash extract was not effective for the removal of K, as the K content of the WBA extract significantly increased the K levels within the system compared with the blank without added WBA extract (Table 7). Our results suggest that carbon availability is the primary driver of algal biomass growth, followed by N and P, which is consistent with the elemental composition of *Chlorella* reported in the literature (CO_{0.48}H_{1.83}N_{0.11}P_{0.01}) or the commonly used Redfield ratio for phytoplankton (C–N–P ratio of 106:16:1) [53,85–87]. During the correlation of the measured biomass increase with the consumption of the measured nutrients, relatively low correlation values were demonstrated (P: R² = 0.5569, K: R² = 0.5755, N: R² = 0.7584), which suggests that N availability affected the algal growth across the treatments. However, other factors probably had a significantly greater impact.

Based on the statistical analysis of the initial algal inoculation and nutrient optimization, significantly higher final algal concentrations, biomass change, and specific growth rates were detected in the different WBA treatments compared with the control. The highest specific growth rates and the largest increase in biomass were observed for the A2N100 and A5N50 treatments ($p < 0.001$) (Table 8). Although the specific growth rate in these treatments ranked only within the top three for net biomass increase, this observation suggests that specific growth rate more effectively reflected the efficiency of each treatment, as a useful metric to optimize and determine the appropriate initial parameters. However, the equation for this derived parameter is fundamentally based on the initial algal quantity and the value obtained is inversely proportional to this initial amount [72]. Consequently, specific growth rates were not sequential in relation to the measured parameters in the various treatments, and thus cannot be used for final decision making.

According to our results, the WBA extract effectively absorbed CO_2 without exhibiting any toxic effects on algal growth. However, at higher concentrations, it inhibited growth, resulting in an extended lag phase. Algae can utilize the CO_2 from the WBA extract as a carbon source, leading to CO_2 depletion, increased biomass, and a rise in pH levels. Furthermore, the ash extract can be partially regenerated, so the pH drops from 12 to 6, and then stabilizes around 10 with 10% treatment. Notably, the highest CO_2 removal efficiency (93%) was also observed for the 10% treatment, with a 29-fold increase in biomass, a specific growth rate of 0.84, and pH restoration all accomplished within a 4-day period. Therefore, the 10% WBA addition treatment was selected for further research.

However, the composition of the WBA can vary due to several factors (e.g., duration of combustion, temperature, and the characteristics of the raw material) [61,62]. Our results indicate that WBA can serve as an effective tool for CO_2 sequestration, eliminating the need for inorganic alkalis such as KOH, $\text{Ca}(\text{OH})_2$, and NaOH in waste management applications. During utilization in flue gas cleaning to capture CO_2 , algae can regenerate the solution from a pH of 6 to around 11, starting from an initial pH of 12. The quantity of initial algal biomass significantly affects the growth rate and pH changes, offering a more precise measure of growth dynamics than optical density. After the uptake of CO_2 from bicarbonate ions (HCO_3^-), residual hydroxide ions (OH^-) remain, which increasingly inhibit algal growth as pH approaches 10. The maximum growth threshold was 273 mg dry weight per liter, probably determined by the amount of carbon sequestered by the ash extract. This relationship remained consistent across different treatment conditions. Based on the elemental composition of *Chlorella*, approximately 50% of the mass of dry biomass consists of carbon, which facilitates a straightforward estimation of the amount of carbon captured from the solution. Our findings relate to a broader environmental context; if the acidification of seas is not caused by CO_2 dissolving in the water, then algae will not be able to restore the pH of the water to neutral through biomass production and the associated release of hydroxide ions. The presence of aquatic pollutants in the oceans can contribute to acidification, not only limiting the ability of oceans to absorb carbon dioxide but also reducing the activity of the algae in these ecosystems.

The growth of algae is primarily limited by carbon availability. However, growth dynamics are also significantly affected by nutrient availability and the initial biomass of the algae. In addition, the quantity of the nutrient solution is crucial for the determination of the growth rate although it does not dictate the maximum growth capacity, as evidenced by the flattening of the growth curves despite the significant residual nutrients. K was present in excess due to the presence of the ash. Thus, the algae could only absorb a fraction of this excess. The measured P indicated higher consumption rates at higher initial concentrations, probably due to the presence of excess N in these scenarios. At low initial concentrations, N was almost completely depleted, while at higher initial values, the rate of consumption was proportionally lower. Nevertheless, this N loss appeared disproportionate when quantified and does not explain the observed increase in biomass calculated by Mandalam and Palsson (6.5% N) [86]. It is probable that while N ultimately limited growth, the algae managed to partially compensate for this deficiency by utilizing other nutrients.

The comparison of experimental results on microalgal growth with the literature data is often challenging. It is very difficult to compare the results of tests performed under varying conditions. For such tests, it is essential to present results in comparable units and standardized metrics. However, the literature lacks a unified approach: some studies report biomass growth per unit area [88–92], while others use per-unit-volume metrics [93–98]. Area-based growth metrics are inherently less comparable as they fail to account for culture depth [99]. As a result, this study utilizes volumetric-specific metrics to express biomass growth. Nevertheless, the objective comparability of results is still limited by the duration of the experiment. Although this may seem manageable using a time-specific metric, such as $\text{mg L}^{-1} \text{ day}^{-1}$, this approach is significantly influenced by the chosen time interval. For example, the sigmoidal growth curves in Figure 5 demonstrate that excluding the saturation phase after day 3 or the initial lag phase could result in significantly higher volumetric

daily productivity values, reflecting only the exponential growth phase. Some authors have reported productivity values for specific days without including average values, potentially resulting in overly optimistic projections when used in literature-based assessments of microalgae's potential for industrial applications. Since our proposed technique could realistically be applied in batch systems at an operational scale, it is crucial to report realistic values reflecting the average productivity over the entire growth cycle.

During the 4-day experiment, the highest volumetric productivity was recorded in the A5N50 and A8N100 treatments, with values of $65.3 \pm 2.2 \text{ mg L}^{-1} \text{ day}^{-1}$ and $65.0 \pm 1.0 \text{ mg L}^{-1} \text{ day}^{-1}$, respectively. These observed values are statistically indistinguishable. The maximum productivity was detected after 50 h (2.08 days) for A8N100 and 62 h (2.58 days) for A5N50, with maximum values of $93.9 \pm 4.5 \text{ mg L}^{-1} \text{ day}^{-1}$ and $79.3 \pm 5.7 \text{ mg L}^{-1} \text{ day}^{-1}$, respectively. Although the duration of the experiment was 4 days, this approach also allowed the determination of the optimal culturing duration under the specific treatment conditions applied. On an annual scale, these results correspond to 34.28 kg and 28.95 kg of dry algal biomass per cubic meter, respectively. High variability was observed in the reported values of biomass productivity, while the reported values for *Chlorella* species range from 2.98 to 5370 $\text{mg L}^{-1} \text{ day}^{-1}$ [100]. According to our prior study, *C. vulgaris* suspensions at maximum density (the determined algal biomass was 770 and 960 mg L^{-1} after filtration and evaporation, respectively) exhibit optical properties that limit precise quantification [58]. Our previous findings suggest that studies reporting values exceeding 1 g L^{-1} based on optical density measurements may include methodological inconsistencies or errors in biomass conversion. The reported values in the present study are consistent with the ranges reported in several studies [95,101], where the range of the observed biomass productivity was 35.83–81.67 $\text{mg L}^{-1} \text{ day}^{-1}$. Our results are also in good agreement with the results of a study where *C. vulgaris* cultured in wastewater with 10% CO_2 injection achieved 381.16–499.52 mg L^{-1} productivity over 23 days, corresponding to daily averages of 16.5–21.7 $\text{mg L}^{-1} \text{ day}^{-1}$ [101]. Additionally, cultures maintained at constant CO_2 levels (0.04%, 3%, 6%, and 9%) showed the highest biomass growth rate (0.37 $\text{g L}^{-1} \text{ day}^{-1}$) for *C. vulgaris* at 3% CO_2 [39].

Several approaches are available for quantifying CO_2 sequestration during various experiments. The most straightforward method derives from the measured decrease in DIC. In the initial model experiment, where the 10% WBA extract treatment was applied, a DIC reduction of $45.2 \pm 7.96 \text{ mg L}^{-1}$ was observed over 4 days (Table 5), corresponding to an average sequestration rate of $41.4 \pm 7.3 \text{ mg CO}_2 \text{ L}^{-1} \text{ day}^{-1}$. This rate is expected to be higher in the second model experiment, where biomass growth was further optimized. Another method calculates CO_2 sequestration based on biomass growth and its carbon content. According to Mandalam and Palsson, the carbon content of *C. vulgaris* is $62 \pm 15\%$ of biomass dry weight [86]. Using this approach, the CO_2 fixation rate for the A8N100 treatment was determined to be $213 \pm 52 \text{ mg CO}_2 \text{ L}^{-1} \text{ day}^{-1}$. In contrast, the reported maximal fixation rates ranged from 4.51–14.26 $\text{mg CO}_2 \text{ L}^{-1} \text{ day}^{-1}$ to 56.26–85.72 $\text{mg CO}_2 \text{ L}^{-1} \text{ day}^{-1}$ under varying conditions [101]. CO_2 fixation rates of 0.93 and 0.69 $\text{g L}^{-1} \text{ day}^{-1}$ were achieved using 2 and 1 $\text{g L}^{-1} \text{ NaHCO}_3$ in 100 mL flasks, respectively [42]. In another study, *Chlorella* sp. cultured in an airlift photobioreactor produced a biomass of 109–264 $\text{mg L}^{-1} \text{ day}^{-1}$ [102]. Using a third approach, biomass productivity can be converted into CO_2 fixation rates based on the literature values [103], resulting in 1.6–2 kg CO_2 fixed per kg (d.m.) year^{-1} . Across the three approaches, annual CO_2 sequestration rates of 15.1 ± 2.7 , 77.9 ± 18.8 , and 54.4–68.6 $\text{kg CO}_2 \text{ m}^{-3} \text{ year}^{-1}$ were calculated. The lower rates observed in the first model experiment were attributed to suboptimal growth conditions. Based on the calculations, our results indicate that scaling up the model experiment to capture 1 ton of CO_2 per year would require 13–66 m^3 of algal culture.

However, the economic feasibility of such systems is unattainable without valorization of the resulting algal biomass. Moreover, the efficiency of carbon sequestration is closely linked to the downstream utilization of the algal biomass, highlighting the necessity of an integrated approach to evaluating the potential of carbon capture and storage. The tech-

nology demonstrated and optimized through the model experiments is uniquely suitable for operation in closed photobioreactor systems. In open pond systems, injected CO₂ gas bubbles through the water medium (which must be shallow for adequate illumination) and escapes into the atmosphere. Consequently, open systems cannot serve as effective CCS techniques [104–107]. Our proposed method employs a sequential approach, where the CO₂ absorption and solution regeneration phases are spatially separated from the carbon sequestration phase involving algal biomass cultivation. As a result, CO₂ diffusion is unnecessary during the cultivation phase, though oxygen produced via photosynthesis must be vented through a valve. This design improves the efficiency of carbon capture by minimizing gas losses and maintaining controlled cultivation conditions.

5. Conclusions

The involvement of algae-based solutions in water resource management and CCS technologies can contribute to climate change mitigation and support sustainable development goals by preserving aquatic ecosystems and ensuring appropriate water quality. Based on our results, DIC can be measured with the use of the OxiTop device. Although the developed method requires further standardization, OxiTop is also suitable for the detection of the dynamics of gas evolution, which could help to distinguish different carbonate forms. Based on the measurements of the DIC content in the different treatments with different concentrations of WBA solutions after bubbling CO₂ through them (compared with freshly boiled distilled water), WBA could potentially be used in CCS technologies to capture CO₂. However, during the possible application of WBA, we have to take into consideration the possible variability of WBA composition, as well as the additional challenges associated with scaling up microalgae culture systems. Our approach also provides an opportunity to utilize agricultural waste, reducing the need for landfilling and supporting the circular transition. Our model experiments showed that CO₂-enriched WBA was not toxic to the growth of algae. Since they play a crucial role in water quality and the carbon cycle of aquatic ecosystems, algae are able to absorb the CO₂ content from the washing solution, effectively regenerating it, which allows it to be re-alkalized and reused for further CO₂ capture. Among the WBA water extracts tested at different concentrations, a 10% *v/v* concentration was optimal for algae. The addition of 10 WBA promoted the fastest growth, restored pH to initial levels, and achieved the highest DIC consumption. During a subsequent experiment, the optimal initial algal density and the amount of nutrients were determined. We observed that starting with a higher density of algae supported rapid growth but did not require the maximum treatment level. Additionally, using half the amount of nutrients recommended by the manufacturer still resulted in robust growth.

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